

arbocartoR package and app: deterministic ODE model for Aedes dynamics and control

Usefull materials and links:

- [arbocartoR page](#)
- [arobocarto page](#)

Tutorial: How to use arbocartoR

This tutorial introduces the **arbocartoR** package, which simulates:

- Spatial and temporal dynamics of *Aedes* mosquito populations under natural and controlled conditions.
- Epidemiological dynamics of arboviruses (*dengue*, *zika*, *chikungunya*) across human and vector populations.

Setup

Install arbocartoR package:

```
In [1]: # remotes::install_gitlab("astre/arbocartoR", host = "https://gitlab.cirad.fr")
```

Load required packages:

```
In [14]: suppressPackageStartupMessages({
  suppressWarnings({
    library(arbocartoR)
    library(data.table)
    library(magrittr)
    library(tidyterra)
    library(ggplot2)
    library(sf)
    library(TDLM)
  })
})

# Please use the RGIS kernel and add conda's bin directory to the front of PATH
Sys.setenv(PATH = paste("/opt/conda/envs/RGIS/bin",
  Sys.getenv("PATH"), sep = ":"))

Sys.setenv(PROJ_LIB = "/opt/conda/envs/RGIS/share/proj")
```

General Data

The package requires two main datasets:

- **parcels**: each row is a node/parcel with attributes (ID: parcel unique identifier, POP: resident population, Kfix: fix/anthropogenic carrying capacity of the parcel, Kvar: variable/rainfall-dependant carying capacity, etc.).
- **meteo**: meteorological data (ID, DATE, rainfall **RR** - in mm, temperature **TP** - in degrees).

The structure and attributes of the data are detailed in the helpers.

```
In [3]: # ? parcels
```

```
In [4]: # ? meteo
```

An example for both datasets is provided in the package:

```
In [2]: data(parcels)
data(meteo)

head(parcels)
head(meteo)
```

A data.table: 6 × 9

ID	NOM_COM	SURF_HA	STATION	DIFF_ALT	Kfix	Kvar	POP	ALT_M
<chr>	<chr>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
060010000	Aiglun	1538.7791	6081001	-707.06636	3874	28952	94	817.93364
060020000	Amirat	1296.6822	6081001	-515.94537	2664	23437	54	1009.05463
060030000	Andon	5435.6842	6081001	-279.32373	15442	98069	648	1245.67627
060040101	Antibes	218.0164	6004009	69.94415	8699	16365	1334	101.94415
060040102	Antibes	119.2201	6004004	116.39238	5470	10067	2447	129.39238
060040103	Antibes	386.6299	6004002	-41.71158	34682	65470	838	33.28842

A data.table: 6 × 7

ID	DATE	RR	TP	ALT_M	LON	LAT
<chr>	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
4006005	2021-01-01	6.2	-1.30	1517.101	989432.8	6355921
4006005	2021-01-02	0.0	-1.05	1517.101	989432.8	6355921
4006005	2021-01-03	0.0	-2.05	1517.101	989432.8	6355921
4006005	2021-01-04	2.2	-3.60	1517.101	989432.8	6355921
4006005	2021-01-05	0.0	-4.50	1517.101	989432.8	6355921
4006005	2021-01-06	0.0	-5.50	1517.101	989432.8	6355921

Select a subset of parcels

Lets use only 4 parcels to run quick simulations.

```
In [3]: parcels <- parcels[startsWith(ID, "06") & (grepl("Gatt", NOM_COM) | grepl("Saint-Jeannet", NOM_COM) | grepl("La Gaude",
```

Run a simple simulation

Select the species of interest, inform the function with your parcels and meteo objects, as well as the temporal boundaries of the simulation. The period must be covered by the meteorological dataset and must last at least a year (for initialization purposes). Note that the vector population dynamics is deterministic, running multiple simulations would produce identical outputs.

```
In [4]: traj <- run_arbocartoR(  
  vector = "Ae. aegypti",  
  parcels = parcels,  
  meteo = meteo,  
  n_sim = 1,  
  start_date = min(meteo$DATE) %>% as.Date,  
  end_date = max(meteo$DATE) %>% as.Date  
)
```

Parameters were based on Ae. aegypti ecology

Simulation will run for 4 parcels from 2021-01-01 to 2022-11-30

No human mobility considered (mMov = NULL)

Generating parameters for all patches can take time, please be patient.

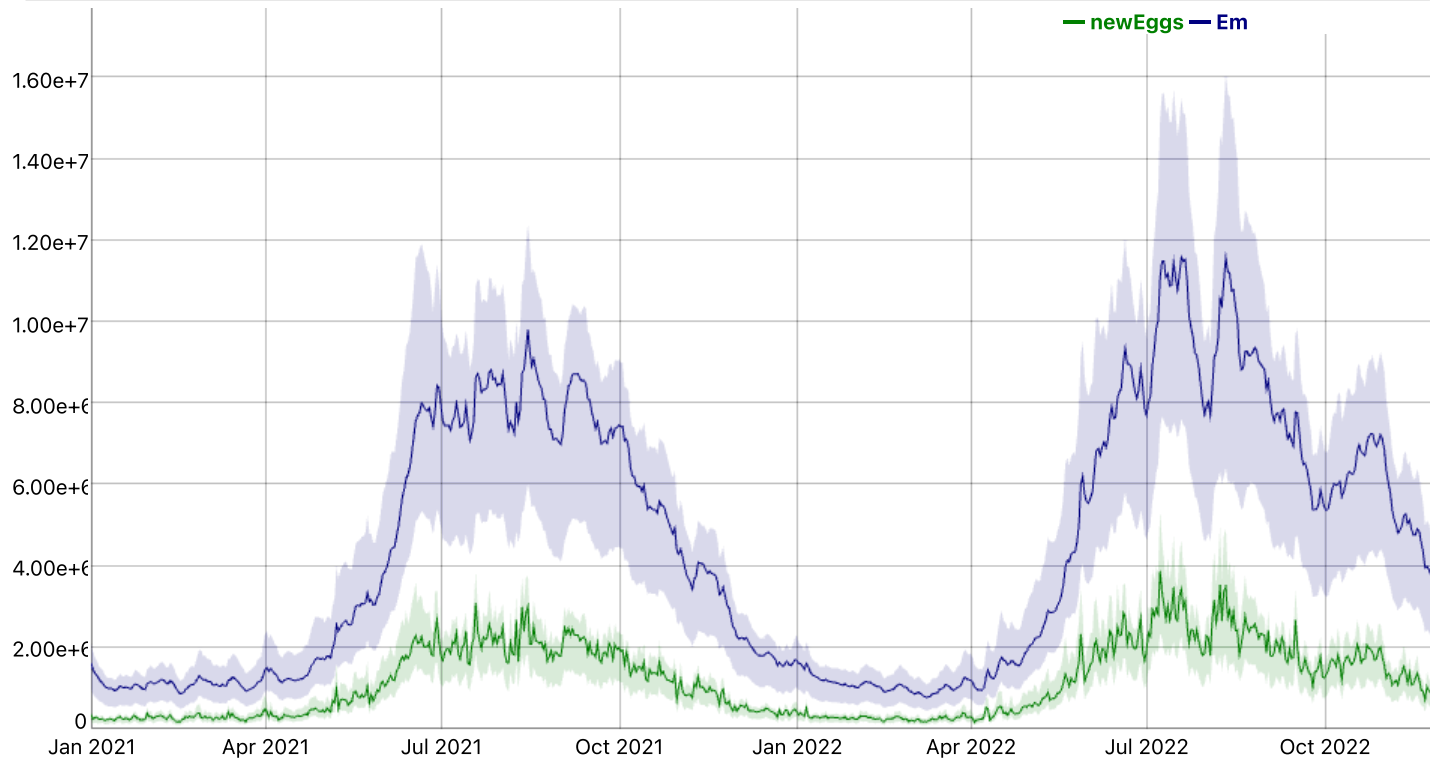
Explore the simulated trajectories compiled as a list of data.table with one data.table per simulation.

```
In [5]: head(traj[[1]])
```

ID	DATE	time	Sh	Em	newEggs	Lm	Pm	Aemm	A1hm	...	A2gm	A2om	R0	interv_Am	z	te
<chr>	<date>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
060640000	2021-01-01	366	4173	910570	174960	555679	80389	4954	2119	...	57934	1330	0	0	1	
060650101	2021-01-01	366	3995	2176470	421320	1217907	187370	12054	5156	...	127440	3221	0	0	1	
060650102	2021-01-01	366	3015	1583608	380580	862468	138864	9262	3692	...	89853	2860	0	0	1	
061220000	2021-01-01	366	4246	1654254	315960	1078820	147412	8725	3744	...	112598	2390	0	0	1	
060640000	2021-01-02	367	4173	807279	122700	564914	78507	4436	1759	...	57331	1496	0	0	1	
060650101	2021-01-02	367	3995	1931970	297420	1223761	182980	10793	4280	...	126260	3634	0	0	1	

Use `plot_TS` function to visualize the trajectories.

```
In [6]: plot_TS(traj, stage = c("newEggs", "Em"))
```



Explore and update parameters

Default parameters used by `arbocartoR` for both *Ae. aegypti* and *Ae. albopictus* are detailed in [Bonnin et al. 2022](#).

The parameters and functions are also described in the helper of the `build_gdata` function.

Explore the default parameters for *aegypti*.

```
In [7]: # ?build_gdata()
gdata_default <- build_gdata(vector = "Ae. aegypti")
```

Parameters were based on *Ae. aegypti* ecology

You can update those parameters using the `build_gdata` function which will build and format the set of ecological and epidemiological parameters required by the model.

Try to update the egg mortality rate (`muE`) and transmission rate from mosquitoes to host (`bmH`) based on *Ae. aegypti* default parameters.

```
In [8]: #Change parameters
gdata_updated <- build_gdata(vector = "Ae. aegypti",
                             bmH = 0.5,
                             muE = 0.001)
```

Parameters were based on *Ae. aegypti* ecology

Compare them:

```
In [9]: gdata_default[c("bMH", "muE", "mu1L")]
gdata_updated[c("bMH", "muE", "mu1L")]
```

\$bMH	0.33
\$muE	0.01
\$mu1L	7e-04
\$bMH	0.5
\$muE	0.001
\$mu1L	7e-04

Run a simulation with those data

```
In [10]: traj <- run_arbocartoR(
  vector = "Ae. aegypti",
  parcels = parcels,
  meteo = meteo,
  gdata = gdata_updated,
  n_sim = 1,
  start_date = min(meteo$DATE) %>% as.Date,
  end_date = max(meteo$DATE) %>% as.Date
)

head(traj[[1]])

plot_TS(traj, stage = "Em")
```

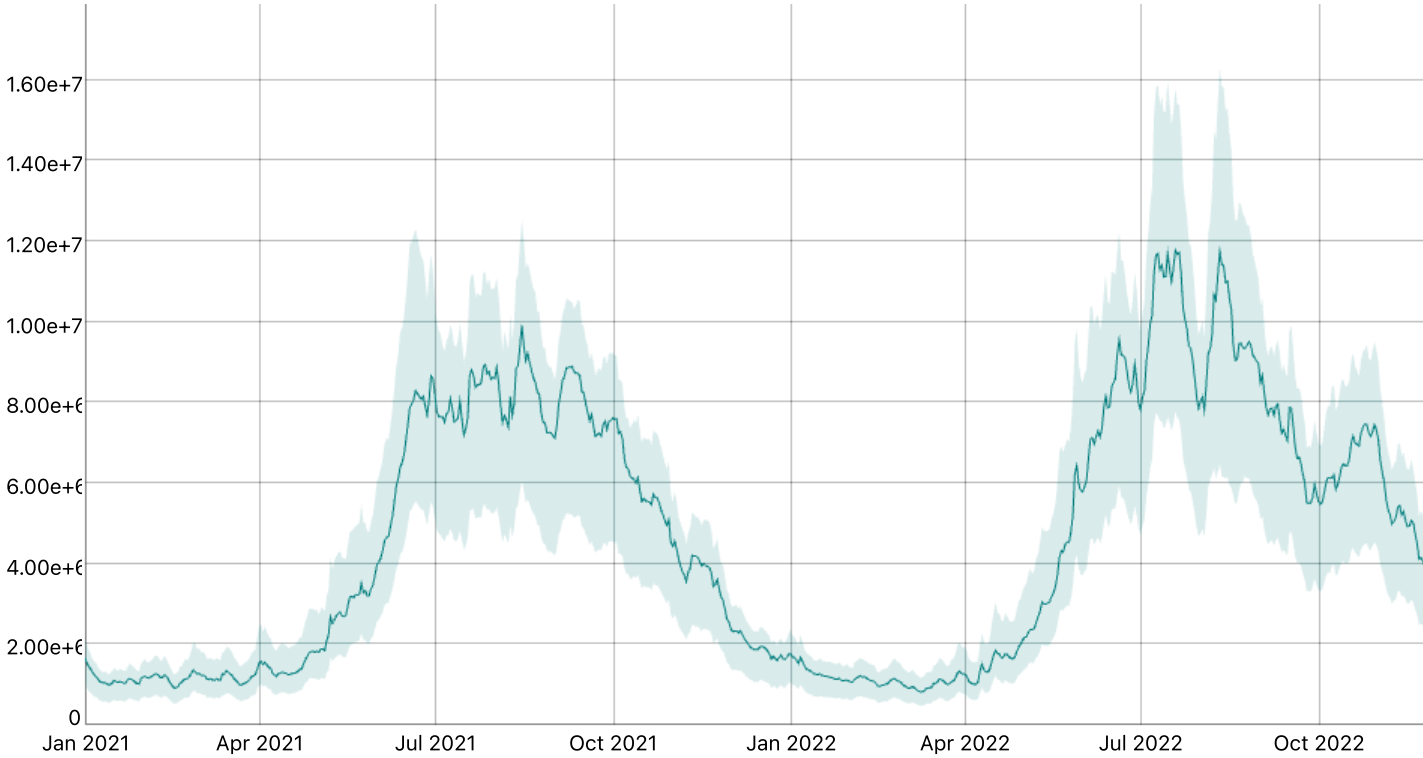
Simulation will run for 4 parcels from 2021-01-01 to 2022-11-30

No human mobility considered (mMov = NULL)

Generating parameters for all patches can take time, please be patient.

A data.table: 6 x 23

ID	DATE	time	Sh	Em	newEggs	Lm	Pm	Aemm	A1hm	...	A2gm	A2om	R0	interv_Am	z	ter
<chr>	<date>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
060640000	2021-01-01	366	4173	948884	175920	567273	82160	5035	2153	...	58085	1333	0	0	1	
060650101	2021-01-01	366	3995	2265663	423360	1242942	191352	12230	5231	...	127677	3226	0	0	1	
060650102	2021-01-01	366	3015	1650168	382200	882101	141760	9388	3740	...	89991	2864	0	0	1	
061220000	2021-01-01	366	4246	1725336	317880	1101608	150724	8876	3809	...	112949	2397	0	0	1	
060640000	2021-01-02	367	4173	845344	123420	577208	80231	4508	1788	...	57488	1500	0	0	1	
060650101	2021-01-02	367	3995	2020786	298860	1249498	186861	10952	4342	...	126513	3641	0	0	1	



Local Preventive Control

Initialize dataset for preventive control actions. Preventive control measures should be described in a data.frame or data.table using the mandatory following columns: 'action', 'loc', 'start', 'end', 'p'.

Define control actions

```
In [11]: prev_control <- data.table(
  action = character(),
  loc = factor(),
  start = structure(numeric(0), class = "Date"),
  end = structure(numeric(0), class = "Date"),
  p = numeric()
)

# Reduce carrying capacity (breeding sites destruction)
prev_control <- rbindlist(list(prev_control, data.table(
  action = "K",
  loc = "060640000",
  start = as.Date("2022/07/01"),
  end = as.Date("2022/07/07"),
  p = 0.2
)), fill = TRUE)

# Increase larvae mortality (larviciding)
prev_control <- rbindlist(list(prev_control, data.table(
  action = "L",
  loc = "060640000",
  start = as.Date("2022/07/01"),
  end = as.Date("2022/07/07"),
  p = 0.2
)), fill = TRUE)

prev_control <- rbindlist(list(prev_control, data.table(
  action = "L",
  loc = "061220000",
  start = as.Date("2022/07/07"),
  end = as.Date("2022/07/15"),
  p = 0.5
)), fill = TRUE)

# Increase adult mortality (fumigation)
prev_control <- rbindlist(list(prev_control, data.table(
  action = "A",
  loc = "061220000",
  start = as.Date("2022/07/01"),
  end = as.Date("2022/07/07"),
  p = 0.2
)), fill = TRUE)

prev_control <- rbindlist(list(prev_control, data.table(
  action = "A",
  loc = "060640000",
  start = as.Date("2022/07/01"),
  end = as.Date("2022/07/07"),
  p = 0.8
)), fill = TRUE)
```

You can also use `build_prev_control()` function to format it using GPS coordinates and buffer width (area covered by the control activities). A `SpatVector` object is required to calculate the spatial intersection between the parcels and the area covered by the control activities.

```
In [15]: SpatVec <- terra::vect("shape/SpatVec.shp")
SpatVec <- SpatVec[SpatVec$ID %in% parcels$ID, ]

prev_control_func <- build_prev_control(action = c("K", "L"),
  lon = 1034406, lat = 6306132,
  start = as.Date("2022/07/01"),
  end = as.Date("2022/07/07"),
  p = c(0.2, 0.8),
  SpatVec = SpatVec,
  buffer_width = 100)
```

Run simulation with preventive control

```
In [16]: traj <- run_arbocartoR(
  vector = "Ae. aegypti",
  parcels = parcels,
  meteo = meteo,
  n_sim = 1,
  start_date = min(meteo$DATE) %>% as.Date,
```

```

end_date = max(meteo$DATE) %>% as.Date,
prev_control = prev_control
)

traj_noControl <- run_arbocartoR(
  vector = "Ae. aegypti",
  parcels = parcels,
  meteo = meteo,
  n_sim = 1,
  start_date = min(meteo$DATE) %>% as.Date,
  end_date = max(meteo$DATE) %>% as.Date
)

prev_control
names(traj[[1]])

traj[[1]]$Lm_nocontrol <- traj_noControl[[1]]$Lm
plot_TS(traj, stage = c("Lm", "Lm_nocontrol"))

traj[[1]]$newEggs_nocontrol <- traj_noControl[[1]]$newEggs
plot_TS(traj, stage = c("newEggs", "newEggs_nocontrol"))

```

Parameters were based on Ae. aegypti ecology

Simulation will run for 4 parcels from 2021-01-01 to 2022-11-30

No human mobility considered (mMov = NULL)

Generating parameters for all patches can take time, please be patient.

Parameters were based on Ae. aegypti ecology

Simulation will run for 4 parcels from 2021-01-01 to 2022-11-30

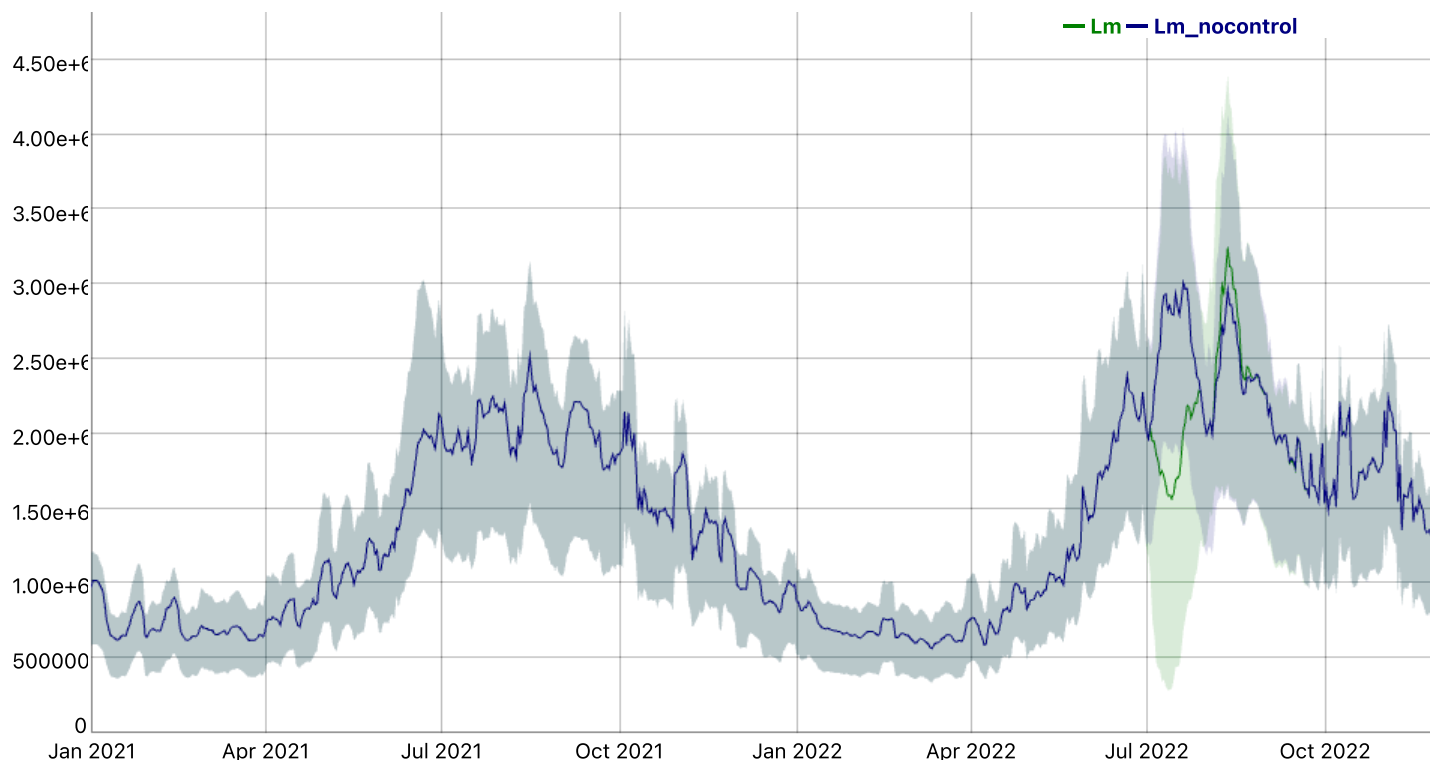
No human mobility considered (mMov = NULL)

Generating parameters for all patches can take time, please be patient.

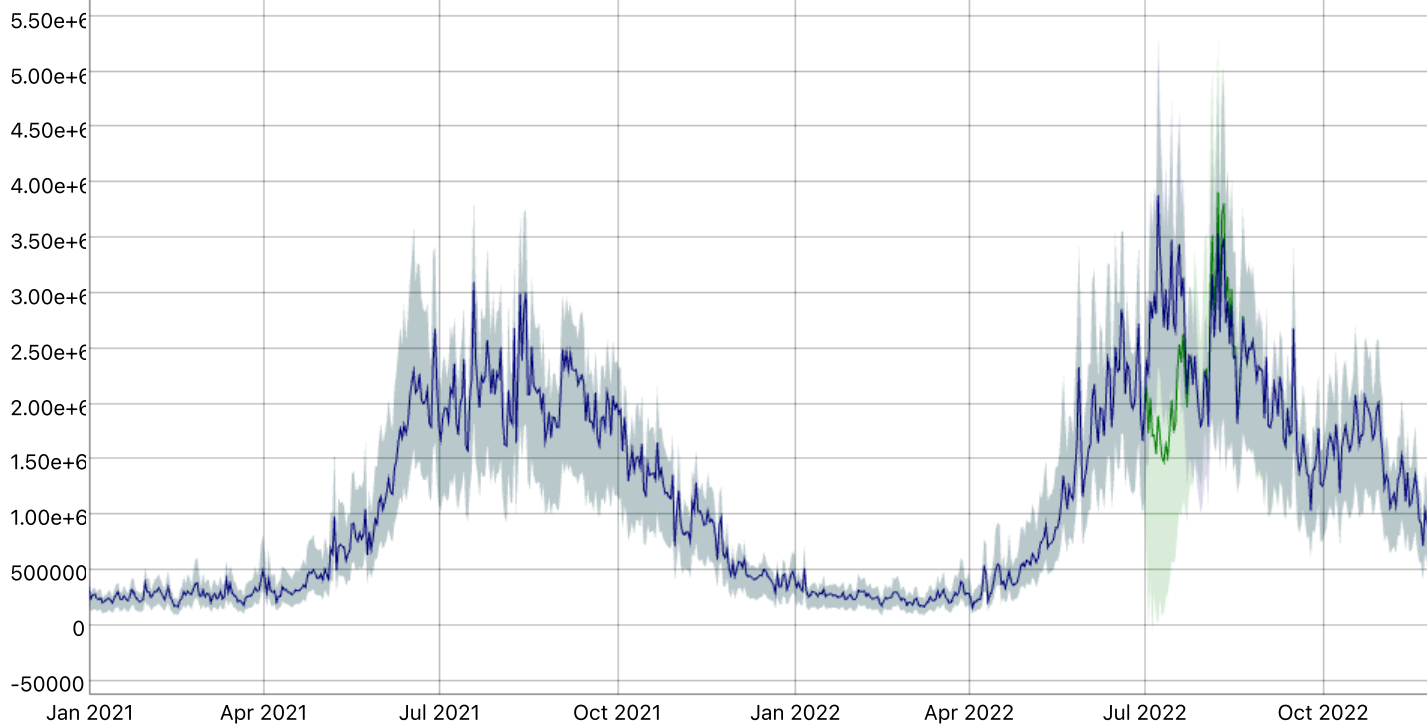
A data.table: 5 × 8

action	loc	start	end	p	start_time	end_time	id
<chr>	<fct>	<date>	<date>	<dbl>	<dbl>	<dbl>	<int>
K	060640000	2022-07-01	2022-07-07	0.2	912	918	1
L	060640000	2022-07-01	2022-07-07	0.2	912	918	2
L	061220000	2022-07-07	2022-07-15	0.5	918	926	3
A	061220000	2022-07-01	2022-07-07	0.2	912	918	4
A	060640000	2022-07-01	2022-07-07	0.8	912	918	5

'ID' · 'DATE' · 'time' · 'Sh' · 'Em' · 'newEggs' · 'Lm' · 'Pm' · 'Aemm' · 'A1hm' · 'A1gm' · 'A1om' · 'A2hm' · 'A2gm' · 'A2om' · 'R0' · 'interv_Am' · 'z' · 'temperature' · 'RR_day' · 'RR_7days' · 'kL' · 'kP'



newEggs newEggs_nocontrol



Introduction of Exposed Individuals

Define an introduction event modifying the epidemiological state of the population at a given time.

```
In [17]: introduction_pts <- build_E_random(  
  period_start = as.Date("2022/08/01"),  
  period_end   = as.Date("2022/09/15"),  
  n_ind = 10,  
  n_events = 1,  
  stage = "Eh",  
  loc = parcels$ID  
)  
  
introduction_pts  
  
traj <- run_arbocartoR(  
  vector = "Ae. aegypti",  
  parcels = parcels,  
  meteo = meteo,  
  n_sim = 5,  
  start_date = as.Date("2022/01/01"),  
  end_date = as.Date("2022/11/30"),  
  introduction_pts = introduction_pts,  
  initMosq = 1e5  
)  
  
plot_TS(traj,  
  stage = c("Eh", "Ih", "Rh", 'A2hmE'),  
  parcels_ids = introduction_pts$dest)
```

A dataframe: 1 × 4

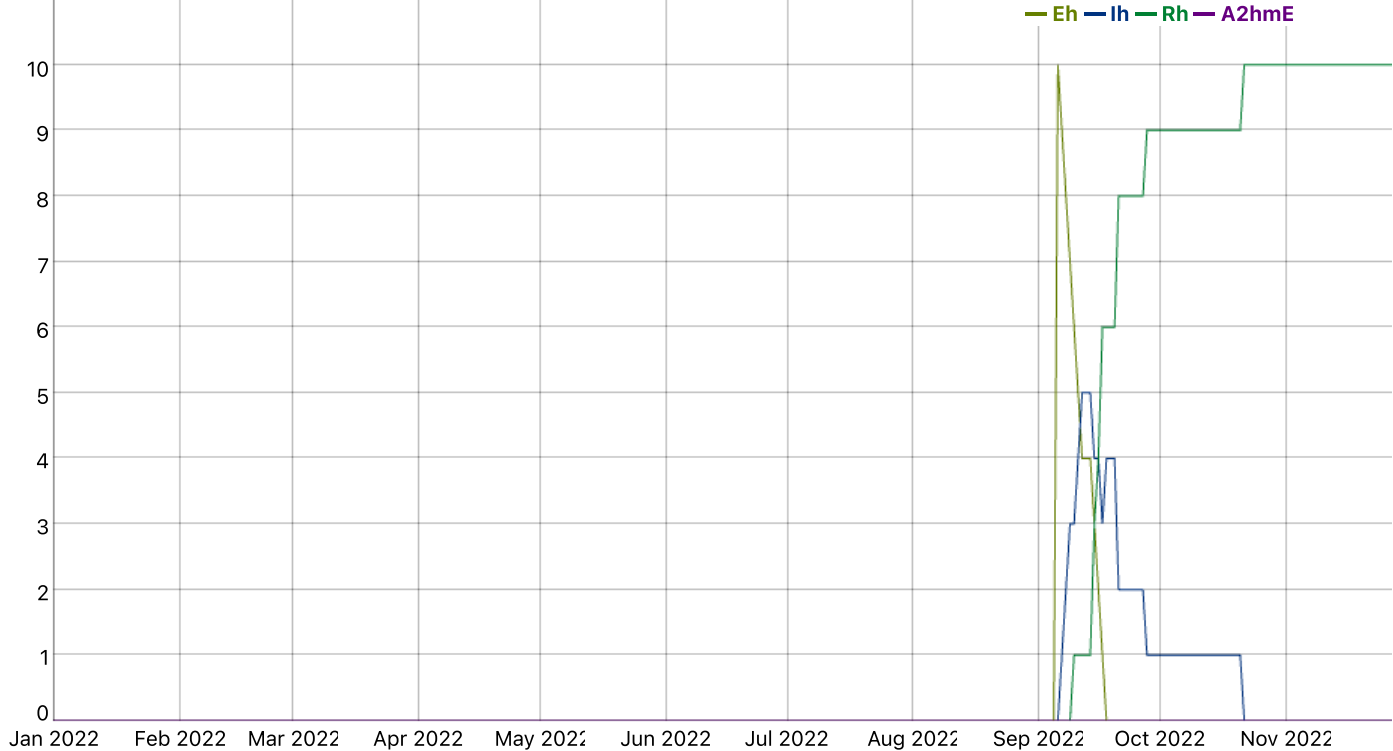
time	dest	n	select
<date>	<fct>	<int>	<fct>
2022-09-06	060640000	10	Eh

Parameters were based on Ae. aegypti ecology

Simulation will run for 4 parcels from 2022-01-01 to 2022-11-30

No human mobility considered (mMov = NULL)

Generating parameters for all patches can take time, please be patient.



Human Mobility

Introduce human mobility between parcels based on the spatial structure of the area of interest, distribution law based on distance between parcels and resident population.

```
In [18]: SpatVec <- terra::vect("shape/SpatVec.shp")
SpatVec <- SpatVec[SpatVec$ID %in% parcels$ID, ]

mMov <- build_mMov(SpatVec,
  law = "NGravExp",
  p2move = 0.715,
  verbose = TRUE)

SpatVec$pMovij <- mMov$proba_ij[rownames(mMov$proba_ij) == "060650102",]
SpatVec$pMovji <- mMov$proba_ji[rownames(mMov$proba_ji) == "060650102",]

ggplot(SpatVec) +
  geom_spatvector(aes(fill = pMovij), color = NA) +
  scale_fill_continuous(low = "white", high = "#590003") +
  ggtitle("Probability to visit a parcel (residents of 061220000)") +
  labs(fill = "")

ggplot(SpatVec) +
  geom_spatvector(aes(fill = pMovji), color = NA) +
  scale_fill_continuous(low = "white", high = "#590003") +
  ggtitle("Probability of visiting 061220000 (depending on residence)") +
  labs(fill = "")
```

Checks

Turn to sf

Build mi, mj, Oi

Compute distance between patches

|

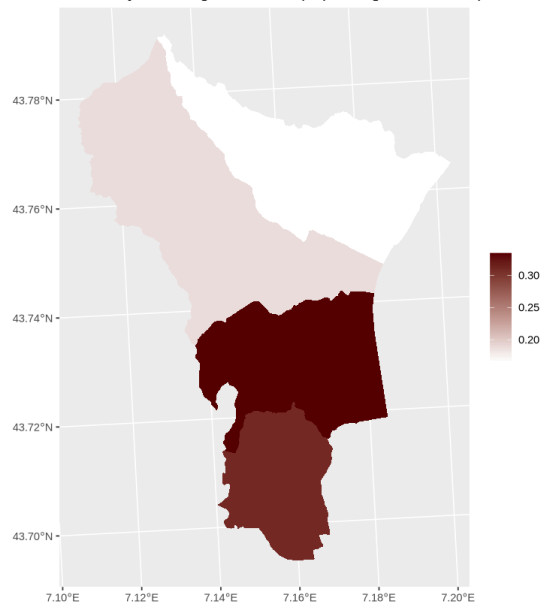
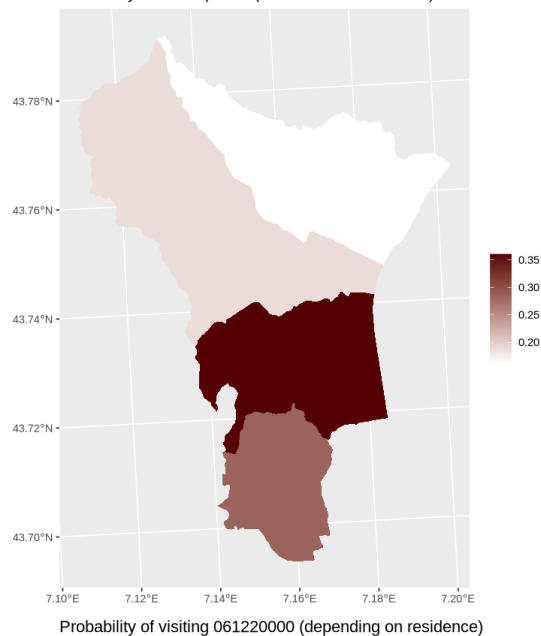
| 0%

The inputs passed the format_and_names checks successfully!

Compute travelling distributions

Normalize probabilities

Probability to visit a parcel (residents of 061220000)



Run the simulation with mobility

```
In [19]: traj_mMov <- run_arbocartoR(
  vector = "Ae. aegypti",
  parcels = parcels,
  meteo = meteo,
  mMov = mMov,
  n_sim = 1,
  start_date = as.Date("2022/01/01"),
  end_date = as.Date("2022/11/30"),
  introduction_pts = introduction_pts,
  initMosq = 1e5
)
```

Parameters were based on Ae. aegypti ecology

Simulation will run for 4 parcels from 2022-01-01 to 2022-11-30

Generating parameters for all patches can take time, please be patient.

Explore and compare the output

```
In [20]: message("Parcels of introduction")
introduction_pts

message("Infected parcels without mobility")
traj[[1]][Eh > 0, unique(ID)]

message("Infected parcels with mobility")
traj_mMov[[1]][Eh > 0, unique(ID)]
```

Parcels of introduction

A data.frame: 1 × 4

time	dest	n	select
<date>	<fct>	<int>	<fct>
2022-09-06	060640000	10	Eh

Infected parcels without mobility

'060640000'

Infected parcels with mobility

'060640000' · '061220000' · '060650101' · '060650102'

In []: